Mohamed Aziz Mrad

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Profile

Mathematics and Computer Science student at Sorbonne Université, with a strong academic record, statistical modelling coursework, and software-development experience. Interested in machine learning, algorithms, and applied mathematical research.

Education

Double Bachelor's Degree in Mathematics and Computer Science 2025 – present
Sorbonne Université, Paris, France

- Currently in second year. First-year annual average: 17.77/20.

Baccalauréat, with honours (*Très Bien*) 2021 – 2025
Lycée Pilote Bourguiba, Tunis, Tunisia

- Overall average: 18.96/20.

Advanced Coursework & Independent Study

- **Machine Learning (Stanford CS229, selected topics):** Supervised learning formulations, Maximum Likelihood Estimation (MLE), gradient descent methods, and binary logistic regression.
- **Design & Analysis of Algorithms (MIT 6.006):** Asymptotic analysis, divide-and-conquer, dynamic programming, and fundamental graph search algorithms.

Research & Technical Projects

Econometric Regression & Diagnostic Modelling R, RStudio
Sorbonne Université, Supervised Research Workshop (Grade: 95/100)

Repository: github.com/Mrad-MedAziz/Baseball-salary-econometrics

- Co-authored an econometric investigation analyzing annual compensation determinants for Major League Baseball hitters ($n = 263$) based on the Hoaglin & Velleman (1995) framework.
- Formulated an Ordinary Least Squares (OLS) multiple regression model combining seasonal performance, average career rate, and career duration, accounting for 62.1% of log-salary variance ($R^2_{\text{adj}} = 0.617$).
- Engineered the normalized metric `Hits_per_year` to decouple cumulative longevity from sustained batting efficiency.
- Evaluated Gauss-Markov assumptions via residual diagnostic plots and Shapiro-Wilk testing ($W = 0.9878$, $p = 0.0248$), identifying non-linear career longevity effects as a primary structural constraint.
- Published a reproducible R script, diagnostic figures, and the complete 11-page research report.

numerical-optimization-ml Python, NumPy
Independent Technical Project

Repository: github.com/Mrad-MedAziz/numerical-optimization-ml

- Implemented core optimization routines and generalized linear models from scratch in pure NumPy without machine learning libraries.
- Derived and programmed the closed-form Normal Equations and batch gradient descent for Ordinary Least Squares regression.
- Developed binary logistic regression using the second-order Newton-Raphson method (Iteratively Reweighted Least Squares).
- Implemented backtracking line search enforcing Armijo conditions to ensure guaranteed descent steps.
- Produced convergence benchmarks illustrating quadratic versus linear convergence properties.

Healthcare AI Concept & Ethics Summer 2024
Summer Innovation Camp, Avila University, Tunis, Tunisia

- Co-designed an assistive clinical tool concept identifying genetic mutation patterns from patient records.
- Structured the algorithmic workflow to present verifiable evidence to physicians, preserving human clinical autonomy and decision authority.

Academic Distinctions and Certifications

- **Double Gold Medal**, National Mathematics Games, Tunisia (2023 and 2024)

- **Mathematics Olympiad distinction** (April 2024)
- **SAT Excellence Certificate**, Amideast Tunis (2024): Mathematics 790/800; Reading and Writing 700/800
- **English C1 Certificate**, Ardmore Language School, United Kingdom (2023)
- **Duolingo English Test**: 135

Technical Skills

- **Programming:** Python; C; R (statistics coursework); Lua.
- **Currently learning:** Java; advanced C topics; set theory, mathematical induction, automata theory.
- **Statistics:** Linear and multiple regression, log transformation, outlier detection, residual diagnostics, hypothesis testing.
- **Mathematics:** Analysis, linear algebra, probability, numerical analysis, graph theory and combinatorics.

Languages

- **Arabic:** Native speaker.
- **French:** Fluent; university studies in French at Sorbonne Université.
- **English:** C1 certified; Duolingo English Test 135; SAT Reading and Writing 700/800.

Experience

Mathematics Teacher 2025 – present
SAM Academy, France

- Teach mathematics to middle and high school students: explain course material, guide exercise work, and prepare students for tests and exams.

National Protocol Officer 2023 – 2024
National Coordination of Interact Clubs, Tunisia

- Ensured compliance with official protocol at national gatherings.
- Coordinated communications and relations with guests of honour and official partners.

Protocol Officer and National Representative 2021 – 2023
Interact Club Saint-Germain, Tunisia

- Organised logistics for club and inter-club events.
- Managed information flows between the club and national bodies, and supervised administrative follow-up and activity reports.

Founder and Administrator, Online Community Server 2021 – 2022
Personal project

- Managed server operations and moderation for an online community.
- Customised server content by editing existing Lua scripts and adjusting object placement, dimensions, and visual design.

Founder and Project Lead, "Braille Menus" Inclusion Initiative 2020 – 2021
Tunisia

- Designed and deployed more than 50 braille menus for international and local restaurant brands.
- Negotiated directly with restaurateurs to make their venues more accessible to visually impaired customers.